MODULE - III

Principles of Polymer Chemistry

Polymer

A polymer is composed of high-molecular mass formed by the combination of large number of small molecules.

The process by which the monomers are transformed into polymers is called polymerization.

Classification of Polymer on the basis of Monomers

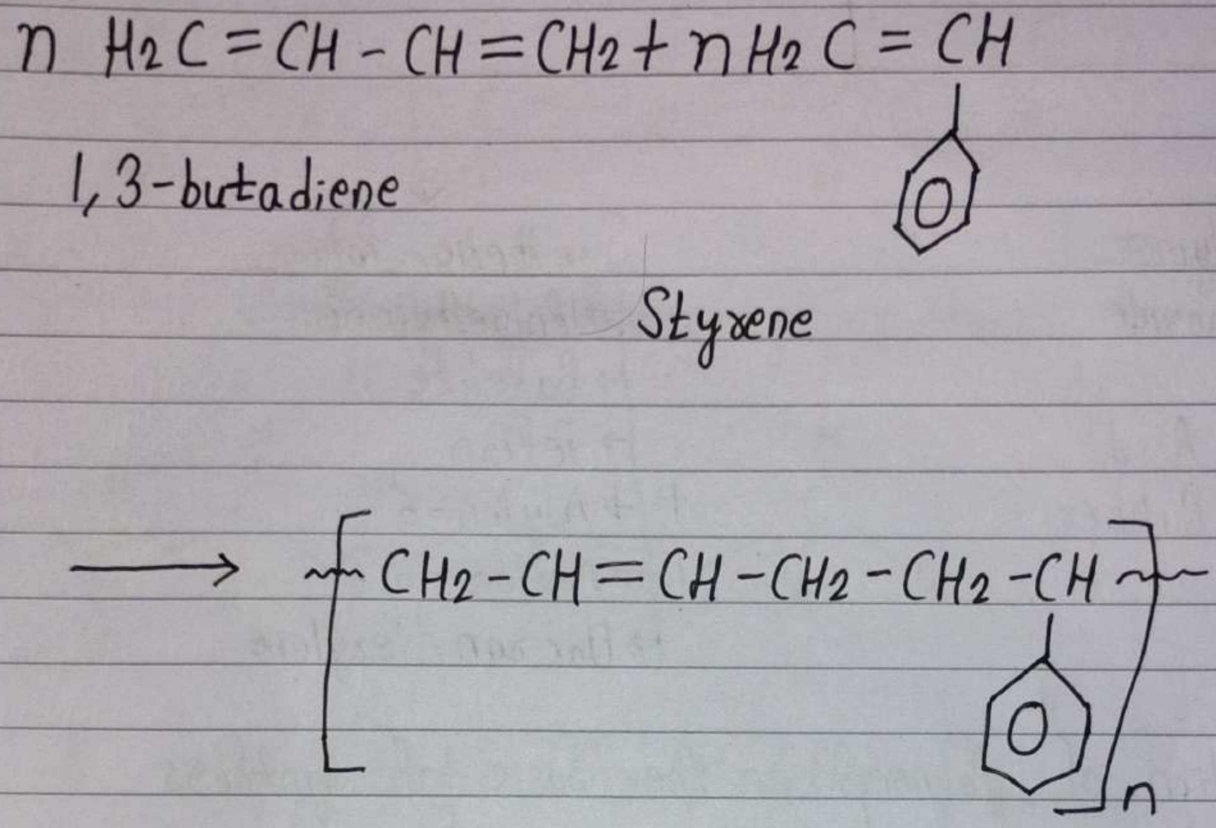
- **Homopolymer** [Monomers = Same]
  - Linear: [A - A - A - A]
  - Branched:
    ```
            A
            |
            A
            |
    A - A - A - A
    ```
  - Cross linked:
    ```
    - A - A - A -
      |   |   |
      A   A   A
      |   |   |
    - A   A   A -
    ```
- **Copolymer** [Monomers = Different]
  - Random Copolymer: [A - A - A - B - A - B]
  - Alternate Copolymer: [A - B - A - B - A - B]
  - *Block Copolymer: [A - A - A - B - B - B]
  - *Graft Copolymer:
    ```
    - A - A - A - A -
              |
              B
              |
              B
              |
              B
    ```

Example of Homopolymer:

$$n\ H_2C = CH_2 \xrightarrow{\text{Polymerisation}} [H_2C - CH_2]_n$$

Ethylene [Monomer] — Polyethylene [Polymer]

Example of Copolymer:



Buna-S

Block Copolymer

Linear copolymers in which the units of each tile formed long continuous sequence (block) are called block copolymer.

Graft Copolymer

Branched copolymer with one kind of monomer in their main chain and another kind of monomer in their side chain.

Classification of Polymer on the basis of Origin

Natural Polymer
- → Polysaccharide
- → Protein
- → Nucleic Acid
- → Natural Rubber

Synthetic Polymer
- → Polyethylene
- → Bakelite
- → Teflon
- → Nylon-6
- → Nylon-6,6
- → Dacron/Terylene

Classification of Polymer on the basis of Synthesis

Addition Polymer

$>C=C</-C\equiv C-$

(No removal)

- → Polyethylene
- → Polypropylene
- → Polyvinyl chloride
- → Buna-S/Buna-N
- → Butyl rubber

Etc.

Condensation Polymer

$-COOH/-NH_2/-COOR/-\overset{O}{\overset{\|}{C}}-NH_2$

$\{-H_2O/HCl/etc\}$

(Group Removal)

→ Nylon-6,6

Monomer: Hexamethylene diamine + Adipic Acid

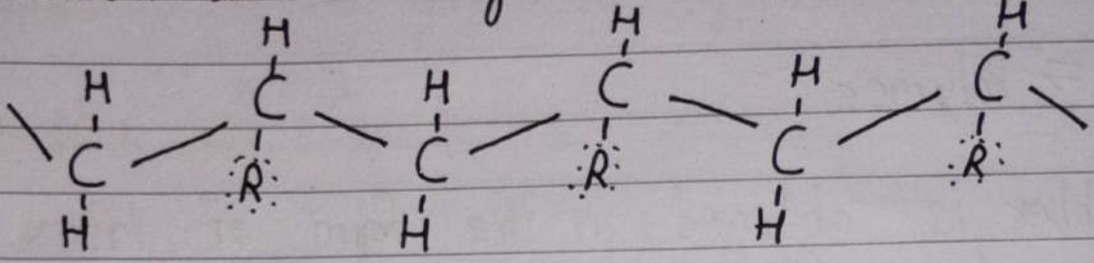
$$n\,H-NH_2-(CH_2)_6-NH-H + n\,HO-\overset{O}{\overset{\|}{C}}-(CH_2)_4-\overset{O}{\overset{\|}{C}}-OH$$

$$\downarrow -nH_2O$$

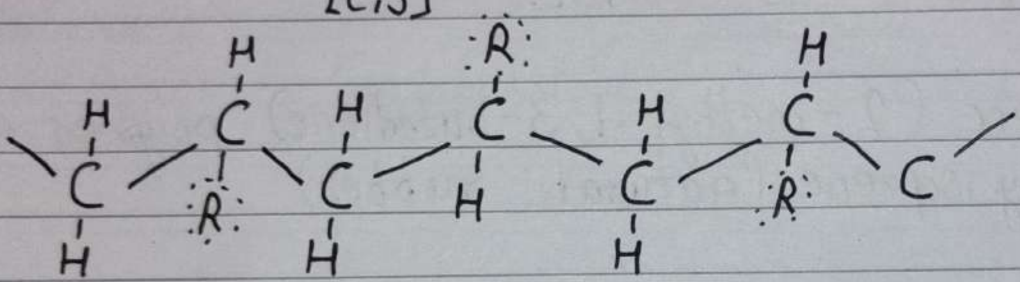
$$\sim\left[NH-(CH_2)_6-NH-\overset{O}{\overset{\|}{C}}-(CH_2)_4-\overset{O}{\overset{\|}{C}}\right]_n\sim$$

(Nylon-6,6)

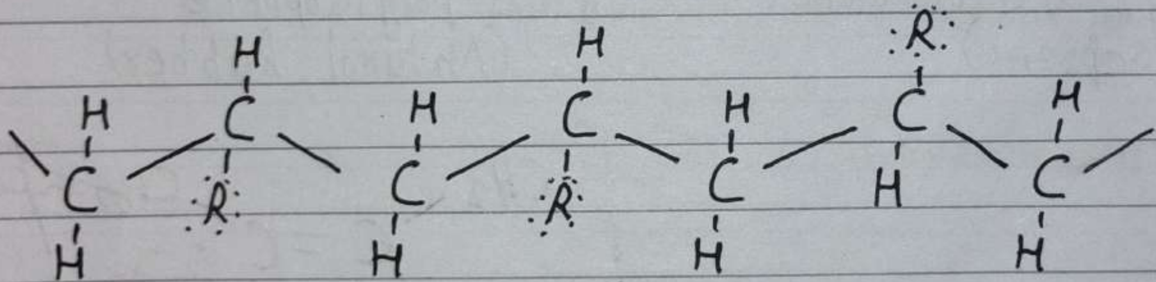
Classissfication of Polymer on the basis of Stereochemistry



Isotactic (Polymer group - same side)
[Cis]



Syndiotactic (Group different)
[Trans]



Atactic (Random)

Natural Rubber

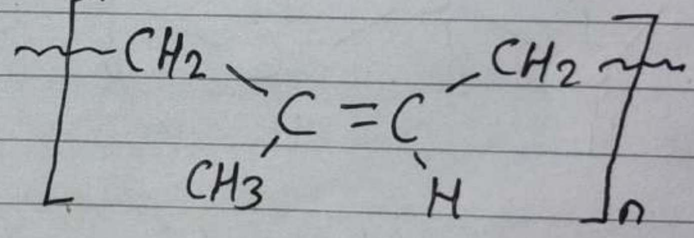
Monomer ⇒ Polymer

Natural rubber is obtained in the form of latex from rubber trees (Haeva Brasiliensis). The latex normally contains 30-65 % rubber. Natural rubber is highly soft and less elastic.

When cis-isoprene (2-methyl 1,3-butadiene) polymerized to form cis-poly isoprene (natural rubber)

$$n\ H_2C=\underset{\displaystyle CH_3}{\underset{|}{C}}-CH=CH_2 \xrightarrow{\text{Polymerization}} \sim\!\!\left[\!-H_2C-\underset{\displaystyle CH_3}{\underset{|}{C}}=CH-CH_2-\right]_n$$

2-methyl 1,3-butadiene (Cis-isoprene) — Cis-polyisoprene (Natural Rubber)



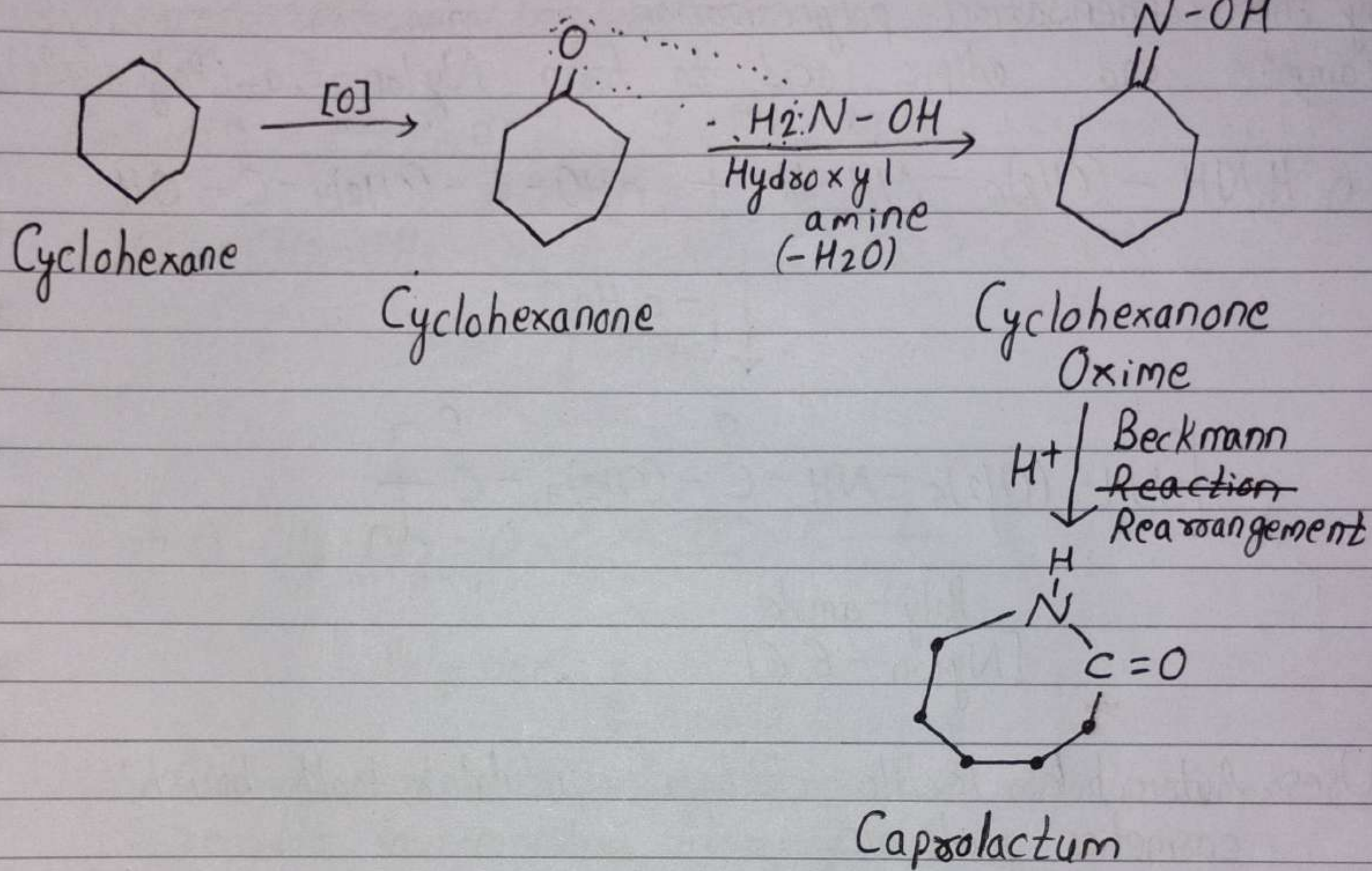
$$\underset{\text{(less elastic)}}{\text{Natural Rubber}} \xrightarrow[\text{[3-5\% Sulphur]}]{\text{Vulcanisation}} \text{Natural rubber (More Elastic)}$$

{Cross-linked with natural rubber}

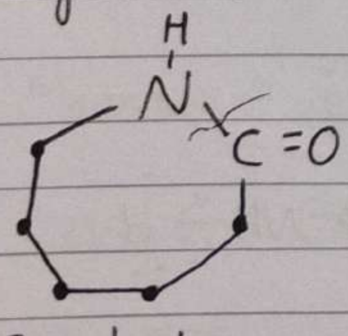
[45 minute]

Synthetic Fibres [Nylon-6 / Nylon-6,6 / Kevlar ⇒ Polymide]

Preparation of Caprolactum:



Nylon-6



Caprolactum
{Monomer}

$\xrightarrow{\text{Hydrolysis}}$ $n\ H - NH - (CH_2)_5 - \overset{O}{\overset{\|}{C}} - OH$

$n\ H - NH - (CH_2)_5 - \overset{O}{\overset{\|}{C}} - OH$

$\downarrow -nH_2O$

$\sim\!\!\left[NH - (CH_2)_5 - \overset{O}{\overset{\|}{C}} - NH - (CH_2)_5 - \overset{O}{\overset{\|}{C}} \right]_n\!\!\sim$

Nylon-6

Nylon-6,6 [Monomers = Hexamethylene diammine + Adipic Acid]

By the condensation polymerization of monomers hexamethylene diammine and adipic acid to form Nylon-6,6 (Poly amide)

$$n\,H{:}NH-(CH_2)_6-NH{:}H + n\,HO{:}\overset{O}{\overset{\|}{C}}-(CH_2)_4-\overset{O}{\overset{\|}{C}}{:}OH$$

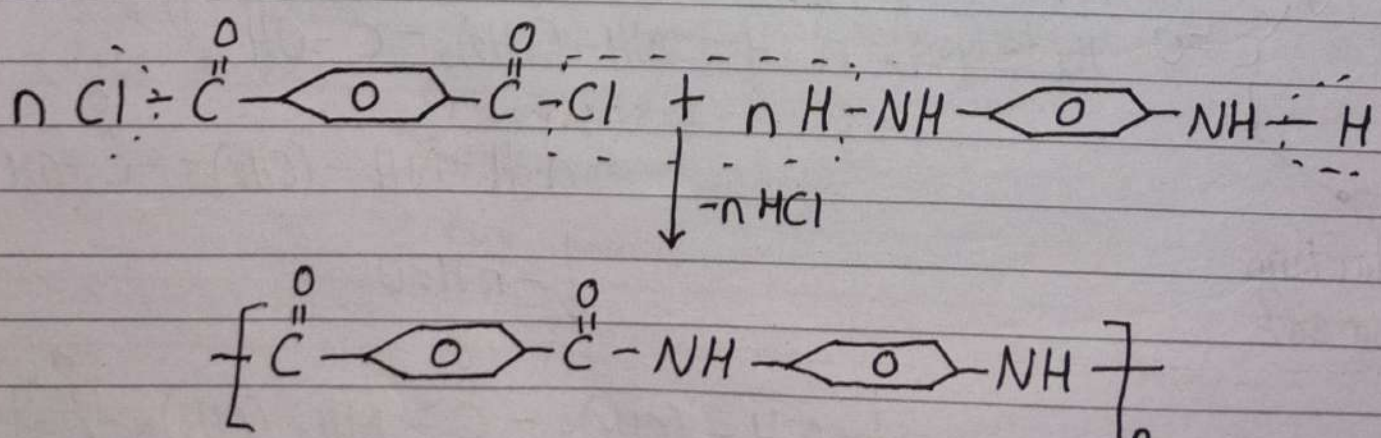
$$\Big\downarrow -n\,H_2O$$

$$\left[NH-(CH_2)_6-NH-\overset{O}{\overset{\|}{C}}-(CH_2)_4-\overset{O}{\overset{\|}{C}}\right]_n$$

Poly amide
[Nylon-6,6]

Uses: Automobiles, textiles, electrical insulator, tooth brush, cosmetics goods, etc.

Kevlar [Monomers = 1,4-dichloro fornyl benzene & 1,4-diaminobenzene]

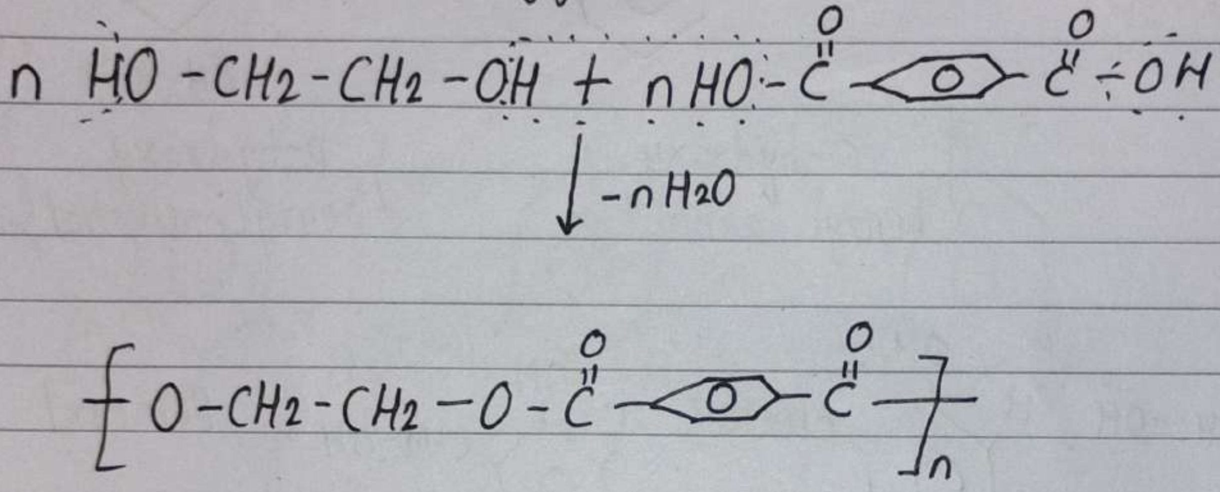


Kevlar

Uses: Bullet-proof jacket, aerospace, aircraft industries, ropes & cables.

Polyester [Dacron/Terylene]

Monomers = Ethylene glycol + Ter-Pthallic acid



Polyester

Uses: Clothing, seat-belts, packing frozen food, magnetic tape recorder, low-moisture absorping power.

PTFE (Polytetrafluoro Ethylene) [Teflon]

Monomer = Tetrafluoro Ethylene

$$n\,F_2C = CF_2 \xrightarrow{\text{Addition}} [F_2C - CF_2]_n$$

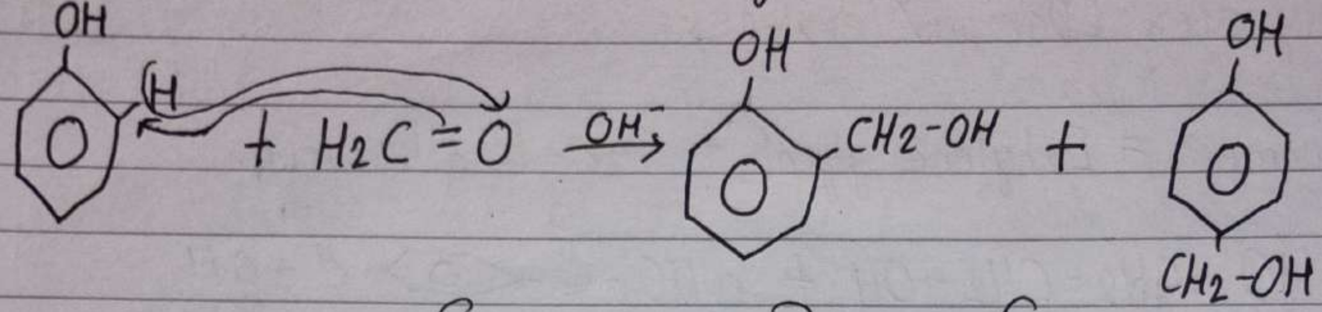
TFE — Teflon

Uses: Non-sticky cookwears, electronics, automobiles, etc.

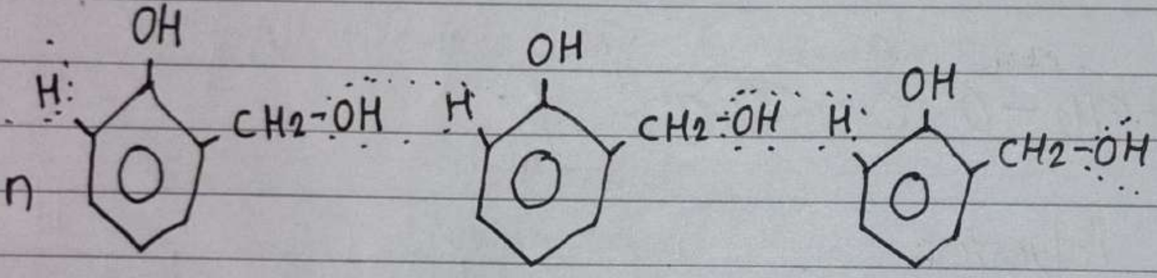
Imp

Bakellite

Monomers = Phenol + Formaldehyde

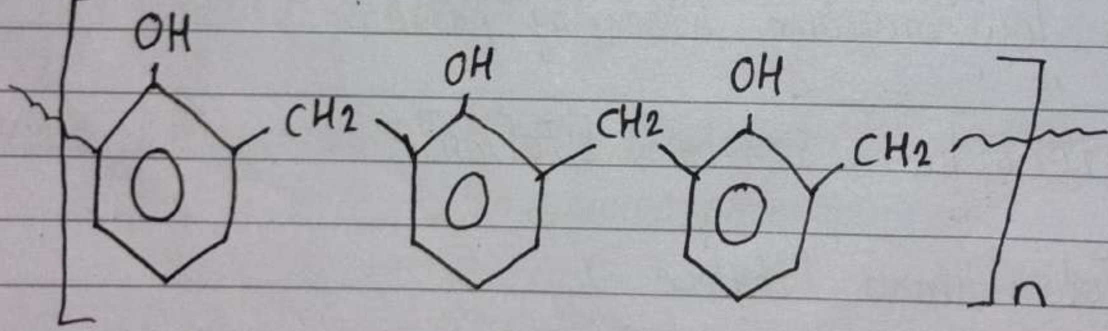


{o-hydroxy benzyl alcohol} {p-hydroxy benzyl alcohol}



[Linear]

↓ Condensation

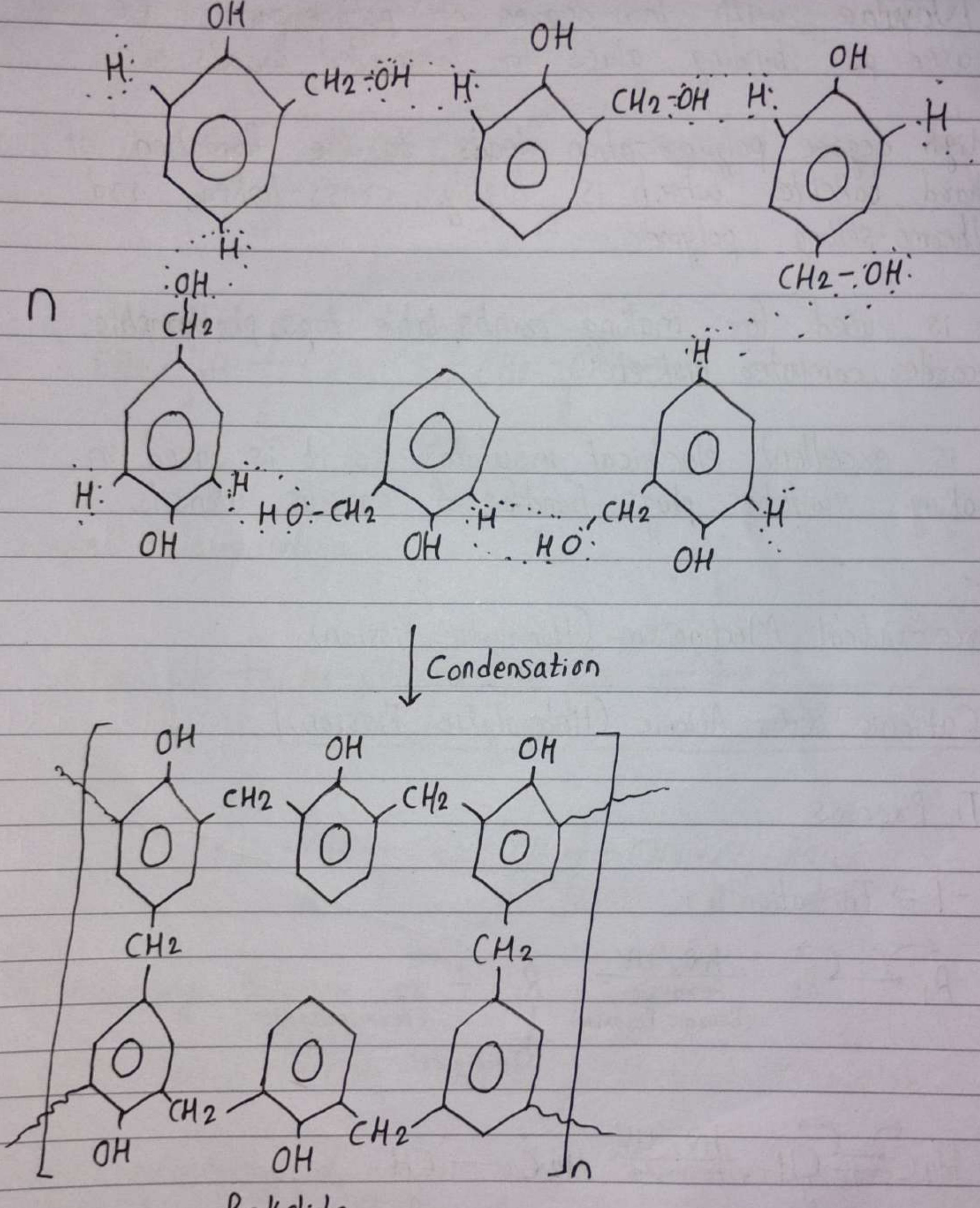


Novalac

→ o & p -hydroxy benzyl alcohol – cross linked = Bakelite

o o p

p o o



Bakelite

When ortho & para hydroxy benzyl alcohol are highly cross linked then bakelite is formed.

~~Us~~ → Novalae with low degree of polymerization are used as binding glues for laminated woods, etc.

→ High degree polymerization leads to the formation of hard bakelite which is highly cross-linked and thermo-setting polymer.

→ It is used for making combs, table tops, photographic recorder, computer disk, etc.

→ It is excellent electrical insulator so it is used in making switches, plugs, handles of various utensils.

Imp

Free-radical Mechanism (Homolytic Fission)

$\boxed{\text{Cationic \& ~~Ca~~ Anionic (Heterolytic Fission)}}$

IPT Process

Step - 1 ⇒ Initiation

$$R_1 - R_2 \xrightarrow[\text{Peroxide}]{h\nu/UV} \underbrace{\dot{R}_1}_{\text{(Initiator)}} + \dot{R}_2$$

(Benzoic Peroxide) (Terminator)

$$H_2C = \underset{\substack{| \\ R}}{CH} \xrightarrow[\text{Peroxide}]{h\nu/UV} H_2\dot{C} - \underset{\substack{| \\ R}}{\dot{C}H}$$

$$\dot{R}_1 \quad H_2\dot{C} - \underset{\substack{| \\ R}}{\dot{C}H} \longrightarrow H_2\overset{\substack{R_1 \\ |}}{C} - \underset{\substack{| \\ R}}{\dot{C}H}$$

Step - 2 ⇒ Propagation

$$R_1-CH_2-\dot{C}H(R) \quad H_2C=CH(R) \quad n\ H_2C=CH(R)$$

$$\downarrow$$

$$R_1-CH_2-CH(R)-\left[CH_2-CH(R)\right]_n-CH_2-\dot{C}H(R)$$

Step - 3 ⇒ Termination

$$R_1-CH_2-CH(R)-\left[CH_2-CH(R)\right]_n-CH_2-\dot{C}H(R) \xrightarrow{R_2}$$

$$R_1-CH_2-CH(R)-\left[CH_2-CH(R)\right]_n-CH_2-CH(R)-R_2$$

Imp

Conducting Polymer

- PTFE/Teflon $\left[CH_2-CH_2\right]_n$ Electrical Insulator

- ~~Na~~ Conductivity of natural rubber (excellent insulator) could be significantly increased by adding carbon black which has natural conductivity.

• In 1930, natural rubber filled with ethyne black was used as an anti-static device in hospitals and other facilities.

Electrically Conducting Polymer

- Intrinsically Conducting Polymer (Extensive Conjugation)
  - Conducting Polymers (having conjugation)
  - Doped Conducting Polymers
- Extrinsically Conducting Polymer (Adding External Substances)
  - Conducting Element Filled Polymer
  - Blended conducting polymer

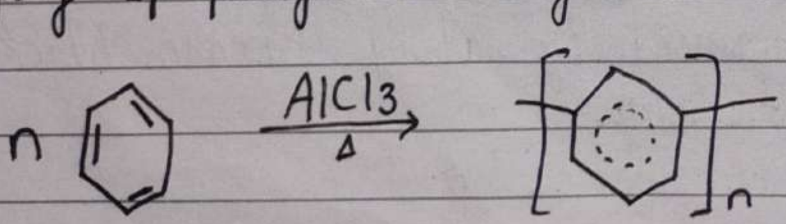
① Conducting Polymers (Having Conjugation)

• Such type of polymers contains conjugated π-electron system.

• These polymers capable of transporting electrical charge in a conjugated π-electron system. i.e alternation of double and single bond.

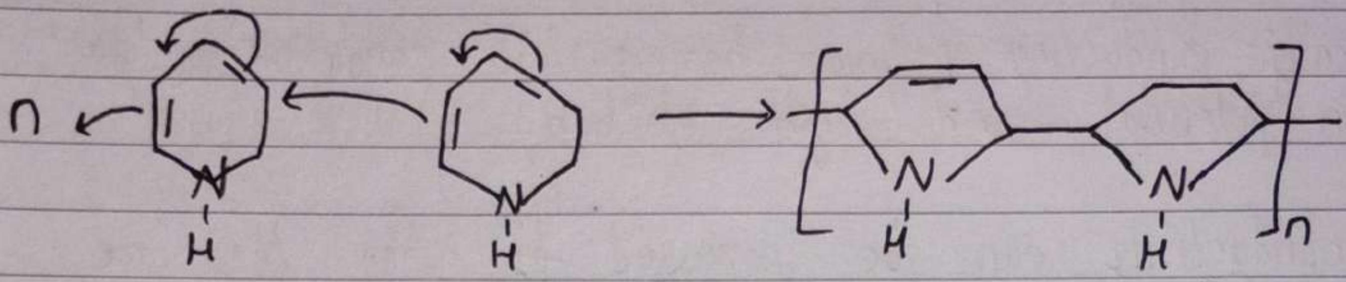
• For example,

(i) poly -[p-phenylene] (Poly benzene)



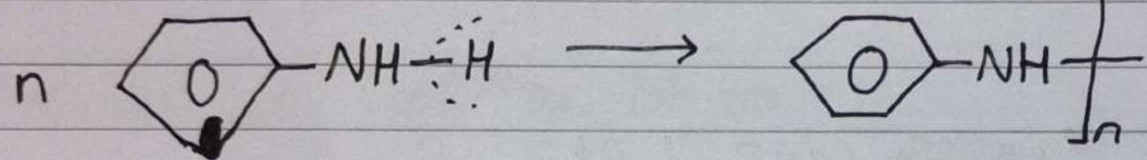
Poly benzene

(ii) Poly pyrrole



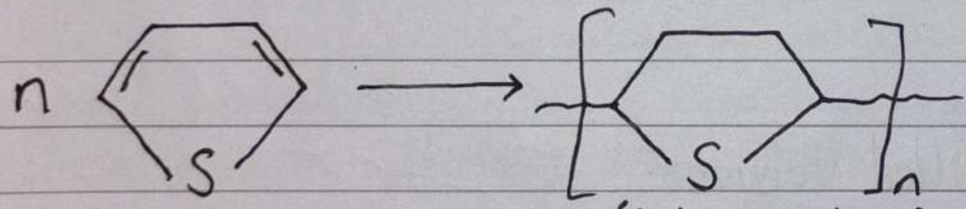
(Pyrrole) (Poly pyrrole)

(iii) Poly ~~thiophene~~ aniline



(Poly aniline)

(iv) Poly thiophene



(Poly thiophene)

② Doped Conducting Polymers

- Intrinsically conducting polymer having low conductivity, low ionization potential and high electron

- These conductivity can be increased ~~by~~ when they are doped with an electron donor such as alkali metals (N-Doping) and electron acceptor such as ~~Ag~~ AsF_5 and iodine (P-Doping).

N-doping

$$(CH)_n + \underset{\substack{\text{Donor} \\ \text{(Alkali metals)}}}{B} \xrightarrow{\text{Reduction}} (CH)_n^- + B^+$$

$$\text{P-doping} + \underset{\substack{\text{Acceptor} \\ (AsF_5 / I_2)}}{B} \xrightarrow{\text{oxidation}} (CH)_n^+ + B^-$$

③ Conducting Element Filled Polymer

- These type of polymers generally have reasonable amount of conductivity, hence the polymer act as binder to hold the conducting material such as carbon black, metal oxide together in solid entity.

- It is low cost, light weight and processed in different shape.

④ Blended Conducting Polymers

- The blending of a convensional polymer with a conducting polymer results in the formation of blended conducting polymers.

- These type of polymer have better physical, electrical, chemical and mechanical properties.

- It is easily processed.

Application of Conducting Polymer:

- It is used in laser, in flat TV.
- Solar cells
- Transistors
- Smart Windows
- LED's
- Carpet Industries
- Electro-magnetic shielding for computers.

Bio-degradable Polymers

- A biodegradable polymer is one that would be decomposed by microorganism.

- Biodegradable polymers degrade mainly by enzymatic hydrolysis and in some cases by oxidation.

- Biodegradable synthetic polymers mostly have functional groups prevalent in proteins, poly-saccharides and lipids.

- PHBV, Poly-glycolic acid, Nylon-2-Nylon-6, Polylactic acid.

- PHBV [Poly β-hydroxy-Butyrate-CO-β-hydroxy valerate]

$$HO-\overset{\overset{CH_3}{|}}{C}H-CH_2-\overset{\overset{O}{\|}}{C}-OH + n\,HO-\overset{\overset{\overset{CH_3}{|}}{CH_2}}{\overset{|}{C}}H-CH_2-COOH$$

(3-hydroxy butanoic acid) (3-hydroxy pentanoic acid)

$$\Big\downarrow -nH_2O$$

$$\sim\!\!\left[O-\overset{\overset{CH_3}{|}}{C}H-CH_2-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{\overset{CH_3}{|}}{CH_2}}{\overset{|}{C}}H-CH_2-\overset{\overset{O}{\|}}{C}\sim\right]_n\!\!\sim$$

(PHBV)

- PHBV is used as packaging material in orthopedic devices and in control drug release.

- $$n\,HO-CH_2-\overset{\overset{O}{\|}}{C}-OH \longrightarrow \left[O-H_2C-\overset{\overset{O}{\|}}{C}\right]_n$$

(Glycolic acid) (Poly glycolic acid)

Used as Rutures

- $$HO-\overset{\overset{CH_3}{|}}{C}H-COOH \longrightarrow \left[O-\overset{\overset{CH_3}{|}}{C}H-\overset{\overset{O}{\|}}{C}\right]_n$$

(Lactic acid) (Polylactic acid)

- Nylon-2-nylon-6

$$\left[NH-CH_2-\overset{\overset{O}{\|}}{C}-NH-(CH_2)_5-\overset{\overset{O}{\|}}{C}\right]_n$$

Molecular Weight of Polymers

- **Number Average Molecular Weight:**

It may be defined as the weight of the sample divided by the number of moles 'n' in the sample.

$$\overline{M}_n = \frac{\text{Weight}}{n} = \frac{n_1M_1 + n_2M_2}{n_1 + n_2}$$

$$\overline{M}_n = \frac{\Sigma n_iM_i}{\Sigma n_i}$$

- **Weight Average Molecular Weight:**

It depends upon the masses of material in different molecular weight fraction. While taking average, molecular wt. of each species is multiplied by the wt. of species not the number.

$$\overline{M}_w = \frac{n_1M_1^2 + n_2M_2^2}{n_1M_1 + n_2M_2}$$

$$\overline{M}_w = \frac{\Sigma n_iM_i^2}{\Sigma n_iM_i}$$

Bioplastic

Bioplastics are plastic materials produced from renewable, biological sources such as vegetable oils, corn starch, sugarcane, wood chips, recycled food waste and lipids.

Bioplastics

- Biologically (Eg: fermentation of sugars or lipids)
- Natural (eg: polysaccharides proteins)
- Chemically (eg: Sugar derivative lipids from either plants or animals.)

- Common plastics such as fossil-fuel plastics (petro-based polymer) are derived from petroleum or natural gas.

- Bioplastics are used for disposable items such as packaging, cutlery and energy related products.

- Bio-based polymers are chemically identical to fossil-fuel based plastics but come from renewable sources.

- Examples; Bio-polyethylene (bio-PE),
  Bio-polyethylene terephthalate (bio-PET),
  Bio-polypropylene,
  & Bio-based nylons.

Common Types of Bioplastic

1. Polysaccharide-based Bioplastics:

(i) Starch-based Plastics: Cheap, abundant and renewable. Usually blended with thermoplastic polymers to make biodegradable and compostable products such as packaging films, carry bags and bubble wrap.

(ii) Cellulose-based Plastics: Derived from cellulose acetate and related compounds. Used in packaging films and coatings. They have good mechanical strength, thermal stability and moisture resistance.

(iii) Other Polysaccharide-based Plastics: Chitosan and alginate can also be processed into plastic forms. Chitosan is compiled of anti-microbial activities and film forming properties which make it biodegradable and determine growth of spoilage.

2. Protein-based Bioplastics:

Made from natural proteins such as gluten and casein. They are biodegradable and used in coatings, films and molded products.

Imp:-

Persistent Organic Pollutants (POPs)

POPs are organic compounds that are resistant to degradation through chemical, biological and photolytic processes.

- They are toxic chemicals that adversely affect human health and environment around the world.

- Because they can be transported by wind, water and soil, most POPs generated in one country or place can affect people and wildlife far from where they are used and released.

- The Stockholm Convention on POPs (2001) [179 Countries], aims to protect human health and the environment by eliminating or reducing their release.

- They include industrial chemicals, pesticides, insecticides, solvents and pharmaceuticals.

- Naturally; from volcanoes.
  Man-made; Aldrin, endrin, HCB, toxaphene, PCBs, DDT, dioxins, mirex, heptachler & dieldrin.

- POPs
  - PBTs (Persistent bioaccumulative and toxic)
  - TOMPs (Toxic Organic Micro Pollutants)

- POPs are highly stable, fat-soluble compounds that accumulate in fatty tissues and move up the food chain.

- They can cause serious health effects such as:- Cancer, Nervous system disorders, Reproductive and immune system damage and Endocrine disruption.

Imp.
Endocrine Disruptors

Endocrine Disruptors (also called hormonally active agents, endocrine disrupting chemicals or endocrine disrupting compounds) are substances that interfere with the body's hormonal system.

These disruptions can cause numerous adverse human health, outcomes including alterations in sperm quality and fertility, abnormalities in sex organs, immune function, cancers, respiratory problems, cardiovascular problems, learning disabilities, elimination of natural hormones in the body that are responsible for human development.

Any system in the body controlled by hormones can be derailed by hormone disruptors.

Thermoplastic	Thermosetting Plastic
• It becomes softer on heating, can be reshaped & reused, soft weak & less brighter, soluble in organic solvent.	• It doesn't become softer on heating, can't be reused, hard & strong, insoluble in organic solvent.
• Ex:- PVC, ABS, PEX, CPVC, etc.	• Ex:- MF, UF, Polyesters, etc.